

(P1) First we should find c , in the way it makes $P_X(i)$ a probability distributions

$$\sum_{i=0}^{\infty} P_X(i) = 1 \quad \Rightarrow \quad c \underbrace{\sum_{i=0}^{\infty} \frac{\lambda^i}{i!}}_{e^{\lambda}} = 1$$

$$c e^{\lambda} = 1$$

$$\Rightarrow c = e^{-\lambda}$$

$$(a) \quad P(X=0) = P_X(0) = \frac{c \lambda^0}{0!} = c = e^{-\lambda}$$

$$(b) \quad P(X > 2) = 1 - P(X \leq 2) = 1 - [P(X=0) + P(X=1) + P(X=2)]$$
$$P(X > 2) = 1 - \left[e^{-\lambda} + e^{-\lambda} \frac{\lambda}{1!} + e^{-\lambda} \frac{\lambda^2}{2!} \right] = 1 - e^{-\lambda} \left[1 + \lambda + \frac{\lambda^2}{2} \right]$$

Problem 2.

$Y =$ The total number of coin tosses until
first head

Y takes values $1, 2, 3, \dots$

for pmf

$P_Y(k) = P(Y=k) \stackrel{?}{=} \text{what question is asking us to find.}$

$Y=k$, it means all the previous trials has resulted
in failure and the k th trial results in success.

$$P_Y(k) = P(Y=k) = (1-p)^{k-1} p \quad k=1, 2, 3, \dots$$

Problem 3.

x	0	1	2	3	4	5	6
$P_X(x)$	0.06	0.13	0.2	0.28	0.25	0.05	0.03

$$(b) \quad P_X(4) = F_X(4^+) - F_X(4^-) = \lim_{x \rightarrow 4^+} F_X(x) - \lim_{x \rightarrow 4^-} F_X(x) = 0.92 - 0.67 = 0.25$$

$$(c) \quad P(1 \leq X \leq 4) = F_X(4) - F_X(1^-) = F_X(4) - \lim_{x \rightarrow 1^-} F_X(x) = 0.92 - 0.06 = 0.86$$

Problem 4.

We need to choose two ranks, one for 3 of a kind, and one for a pair.
once we choose the rank, for each rank there are four suits available,
we need 3 of a kind, so we need to choose 3 suits from 4 suits, we need to
choose 2 suits from 4 suits for the pair.

$$p(\text{Full house}) = \frac{\overbrace{\binom{13}{1}}^{2 \text{ points}} \overbrace{\binom{12}{1}}^{2 \text{ points}} \overbrace{\binom{4}{3}}^{2 \text{ points}} \overbrace{\binom{4}{2}}^{2 \text{ points}}}{\underbrace{\binom{52}{5}}_{2 \text{ points}}}$$

2 points

Problem 5. E = At least one success

E^c = All the trials fail.

n = total number of trials

In each trial

$\nearrow p$	success
$\searrow 1-p$	failure

$$P(E^c) \stackrel{\nearrow n}{=} (\text{probability of failure})^n$$

because trials are independent.

$$P(E^c) = (1-p)^n$$

$$P(E) = 1 - (1-p)^n.$$

(Part b) number of patterns in which we have k success in n trials is $\binom{n}{k}$, in each pattern we have k success and $n-k$ failure, hence,

$$P(\text{exactly } k \text{ success in } n \text{ trials}) = \binom{n}{k} p^k (1-p)^{n-k}$$

Problem 6.

H = Being healthy

H^c = Having the illness.

D = Detecting the illness = That's the test turns positive.

$$P(H) = P(H^c) = 0.5$$

$$P(\text{Detecting the illness} \mid H^c) = 0.98$$

$$P(\text{Detecting the illness} \mid H) = 0.06$$

question is asking for $P(H \mid \text{Detecting the illness}) = P(H \mid D) \stackrel{?}{=}$

we will use Bay's rule:

$$P(H|D) = \frac{P(D|H) P(H)}{P(D)}$$

$$P(D) \stackrel{\uparrow}{=} P(D|H) P(H) + P(D|H^c) P(H^c)$$

law of total probability

$$P(D) = 0.06 \cdot 0.5 + 0.98 \cdot 0.5 = 0.03 + 0.49 = 0.52$$

$$P(H|D) = \frac{0.03}{0.52} = \frac{3}{52}$$

Problem 7.

(a) we have 15 objects, for each object there are three kids

$$\underbrace{3 \quad 3 \quad 3 \quad \dots \quad 3}_{15 \text{ times}} = 3^{15}$$

(b) $\binom{6}{2}\binom{9}{3}$ will be given to the first kid, but there will be 4 snacks left and 6 toys

left that can be distributed with no restriction between second kid and third kid, hence similar to part (a) this can be done in 2^{10} ways, hence.

$$\binom{6}{2}\binom{9}{3} 2^{10}$$

(c) $\underbrace{\binom{6}{2}\binom{9}{3}}_{\substack{\uparrow \\ \text{goes to first} \\ \text{kid}}} \cdot \underbrace{\binom{4}{2}\binom{6}{3}}_{\substack{\downarrow \\ \text{goes to} \\ \text{2nd kid}}} \cdot \underbrace{1}_{\substack{\downarrow \\ \text{whatever is} \\ \text{left goes to} \\ \text{3rd kid}}}$

P8. (a) we will choose 13 card to give to the 1st person, then from 39 cards left we choose 13 card to give to the second person, then from 26 cards left we choose 13 to give to the 3rd person and whatever left goes to the last person.

$$\binom{52}{13} \binom{39}{13} \binom{26}{13}$$

(b) we have 4 queen in total, we will choose 2 queen to give it to the first person and the other 2 queen will be given to the second person. Now first person and second person each needs 11 cards and 3rd and 4th person each needs 13 cards and we should divide rest of 48 cards between them.

$$|E| = \binom{4}{2} \binom{48}{11} \binom{37}{11} \binom{26}{13}$$

$$P(E) = \frac{\binom{4}{2} \binom{48}{11} \binom{37}{11} \binom{26}{13}}{\binom{52}{13} \binom{39}{13} \binom{26}{13}}$$

(c)

$$\frac{\overbrace{\binom{2}{1} \binom{24}{12}}^{\text{1st player}} \overbrace{\binom{39}{13}}^{\text{2nd player}} \overbrace{\binom{26}{13}}^{\text{3rd player}}}{\binom{52}{13} \binom{39}{13} \binom{26}{13}}$$

Problem 9.

A = Box 1 is chosen

A
 $\{1, 2, \dots, 6\}$

B

$\{1, 2, \dots, 10\}$

B = Box 2 is chosen

M = maximum number of Balls chosen is 6

$$P(M|A) = \frac{\binom{5}{3}}{\binom{6}{4}} = \frac{10}{15} = \frac{2}{3} \quad P(M|B) = \frac{\binom{5}{3}}{\binom{10}{4}} = \frac{10}{210} = \frac{1}{21}$$

we will use total probability law

$$(b) \quad P(M) = P(M|A)P(A) + P(M|B)P(B) =$$

$$= \frac{2}{3} \cdot \frac{1}{2} + \frac{1}{21} \cdot \frac{1}{2} = \frac{14 + 1}{42} = \frac{15}{42} = \frac{5}{14}$$

$$(c) \quad P(A|M) = \frac{P(M|A)P(A)}{P(M)} = \frac{\frac{2}{3} \cdot \frac{1}{2}}{\frac{5}{14}} = \frac{\frac{1}{3}}{\frac{5}{14}} = \frac{14}{15}$$

Problem 10.

B = 2 aces are chosen at random from 52 card.

E = Ace of spade is chosen when 2 cards are chosen at random.

F = At least one ace is chosen

For E we know one card is Ace of spades. For the second card 51 possibilities are left.

For $B \cap E$, we know one Ace is Ace-spade, second Ace 3 options are left.

$$P(B|E) = \frac{P(B \cap E)}{P(E)} = \frac{|B \cap E|}{|E|} = \frac{3}{51}$$

$$(b) \quad F \cap B = B \quad \Rightarrow \quad |B| = \binom{4}{2}$$

$$F^c = \text{no aces to be chosen} \quad |F^c| = \binom{48}{2}$$

$$|S| = \binom{52}{2}$$

$$|F| = |S| - |F^c| = \binom{52}{2} - \binom{48}{2}$$

$$P(B|F) = \frac{|B \cap F|}{|F|} = \frac{\binom{4}{2}}{\binom{52}{2} - \binom{48}{2}} = \frac{4 \cdot 3}{52 \cdot 51 - 48 \cdot 47}$$

$$= \frac{12}{2652 - 2256} = \frac{12}{396} = \frac{1}{33}$$

Problem 11.

5 blue
6 red
7 black

$$(a) \quad p(\text{Both Blue}) = \frac{5}{18} \cdot \frac{4}{17} = \frac{20}{306}$$

$$(b) \quad p(\text{different colors}) = 1 - p(\text{Both blue}) - p(\text{both red}) \\ - p(\text{Both black})$$

$$= 1 - \left[\frac{5}{18} \cdot \frac{4}{17} + \frac{6}{18} \cdot \frac{5}{17} + \frac{7 \cdot 6}{18 \cdot 17} \right]$$

$$= 1 - \left[\frac{20 + 30 + 42}{18 \cdot 17} \right] = 1 - \frac{92}{306} = \frac{214}{306}$$

problem 12. $D = \text{own a dog}$
 $C = \text{own a cat}$

$$P(D) = 0.36$$

$$P(C) = 0.3$$

$$P(C|D) = 0.22$$

$$(a) \quad P(C \cap D) = P(C|D) \cdot P(D) = 0.22 \cdot 0.36 = 0.0792$$

$$(b) \quad P(D|C) = \frac{P(C \cap D)}{P(C)} = \frac{0.0792}{0.3} = 0.264$$

Problem 13.
$$D = \overset{H_1}{(A \cap B^c \cap C^c)} \cup \overset{H_2}{(A^c \cap B \cap C^c)} \cup \overset{H_3}{(A^c \cap B^c \cap C)}$$

H_1, H_2 , and H_3 are disjoint hence,

$$P(D) = P(A \cap B^c \cap C^c) + P(A^c \cap B \cap C^c) + P(A^c \cap B^c \cap C)$$

$$P(H_1) = P(A \cap B^c \cap C^c) \underset{\substack{\uparrow \\ \text{independence}}}{=} P(A) \cdot P(B^c) \cdot P(C^c) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{4}$$

$$P(H_2) = P(A^c \cap B \cap C^c) \underset{\substack{\uparrow \\ \text{independence}}}{=} P(A^c) \cdot P(B) \cdot P(C^c) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4} = \frac{1}{8}$$

$$P(H_3) = P(A^c \cap B^c \cap C) = P(A^c) \cdot P(B^c) \cdot P(C) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} = \frac{1}{12}$$

$$(a) \quad P(D) = P(H_1) + P(H_2) + P(H_3) = \frac{1}{4} + \frac{1}{8} + \frac{1}{12} = \frac{6+3+2}{24} = \frac{11}{24}$$

$$(b) \quad P(A|D) = \frac{P(A \cap D)}{P(D)} \quad A \cap D = H_1$$

$$P(A|D) = \frac{P(H_1)}{P(D)} = \frac{\frac{1}{4}}{\frac{11}{24}} = \frac{6}{11}$$

$$P(B|D) = \frac{P(B \cap D)}{P(D)}$$

$$B \cap D = H_2$$

$$\Rightarrow P(B|D) = \frac{P(H_2)}{P(D)} = \frac{\frac{1}{8}}{\frac{11}{24}} = \frac{3}{11}$$

$$P(C|D) = \frac{P(C \cap D)}{P(D)}$$

$$C \cap D = H_3$$

$$P(C|D) = \frac{P(H_3)}{P(D)} = \frac{\frac{1}{12}}{\frac{11}{24}} = \frac{2}{11}$$

$$(c) \quad P(A|B^c) = \frac{P(A \cap (B^c))}{P(B^c)} = \frac{P(A) - P(A \cap B)}{P(B^c)}$$

$$P(A|B \cup C) = \frac{P(A) - P(H_1)}{P(B) + P(C) - P(B) \cdot P(C)} = \frac{\frac{1}{2} - \frac{1}{4}}{\frac{1}{3} + \frac{1}{4} - \frac{1}{12}} = \frac{\frac{1}{4}}{\frac{6}{12}}$$

$$= \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{2}{4} = \frac{1}{2} \quad \checkmark$$

we know A & $B \cup C$ are independent, Hence $P(A|B \cup C) = P(A)$

Problem 14. (a) $\int_{-\infty}^{\infty} P(t) dt = 1 \Rightarrow \int_0^k \sin t dt = 1$

$$-\cos t \Big|_0^k = 1$$

$$-\cos k + 1 = 1 \Rightarrow \cos k = 0$$

$$\Rightarrow k = \frac{\pi}{2}$$

For $0 < t < \frac{\pi}{2}$

$$F(t) = \int_0^t \sin x \, dx = -\cos x \Big|_0^t = 1 - \cos t$$

For $t > \frac{\pi}{2}$

$$F(t) = \int_{-\infty}^t f(t) \, dt = \int_{-\infty}^0 \cancel{f(t) \, dt} + \int_0^{\frac{\pi}{2}} f(t) \, dt + \int_{\frac{\pi}{2}}^t \cancel{f(t) \, dt} = \int_0^{\frac{\pi}{2}} f(t) \, dt = 1$$

$$F(t) = \int_{-\infty}^t f(x) \, dx = \begin{cases} 0 & \text{if } t < 0 \\ 1 - \cos(t) & 0 \leq t \leq \frac{\pi}{2} \\ 1 & t > \frac{\pi}{2} \end{cases}$$

Problem 15.

X takes values $0, 1, 2, 3$

Y takes values $0, 1, 4$

$$P(Y=0) = P(X=1) = 0.4$$

$$P(Y=1) = P(X=0 \text{ or } X=2) = P(X=0) + P(X=2) = 0.2 + 0.1 = 0.3$$

$$P(Y=4) = P(X=3) = 0.3$$

Y	0	1	4
$P_Y(y)$	0.4	0.3	0.3

$$F(y) = \begin{cases} 0 & y < 0 \\ P_Y(0) = 0.4 & 0 \leq y < 1 \\ P_Y(0) + P_Y(1) = 0.7 & 1 \leq y < 4 \\ 1 & 4 \leq y \end{cases}$$